College ranking

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read data

```
library(tidyverse)
cr <- read.csv(file = 'data-BIe5s.csv')
```

Look at names and select some columns

```
names(cr)[1] <- 'Rank'
schools <- cr$School
ranks <- cr$Rank
cr_bloomberg <- cr %>%
  select(names(cr)[c(3,4,5,6,7,8)])
```

Weights from Bloomberg

```
w = c(0.357, 0.258, 0.178, 0.12, 0.086)
cr_bloomberg <- cr_bloomberg %>%
  mutate(my_overall_score = w[1]*Compen..sation + w[2]*Learning + w[3]*Network..ing
         + w[4]*Entrepre..neurship + w[5]*Diversity) %>%
  mutate(my_overall_scaled = 100*(my_overall_score - min(my_overall_score))/(max(my_overall_score)-min(r
  mutate(my_overall_scaled_main = round(my_overall_scaled))
```

Check it witj Ajain's work

```
cr$Scaled.overall.score.with.BBW.weights == cr_bloomberg$my_overall_scaled_main
```

```
##  [1]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [25]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [37]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE
## [49]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [61]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [73]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
```

create my own ranking according to BB weights

```
cr_bloomberg <-  cr_bloomberg %>%
  mutate(my_rank_with_BB_weights = rank(-my_overall_scaled_main, ties.method = 'min'),
         official_ranking = ranks)
```

lets get wrights from MLR

```
m <- lm(formula = Overall.score ~ Compen..sation + Learning +
                                  Network..ing + Entrepre..neurship + Diversity - 1, data = cr_bloomberg
```

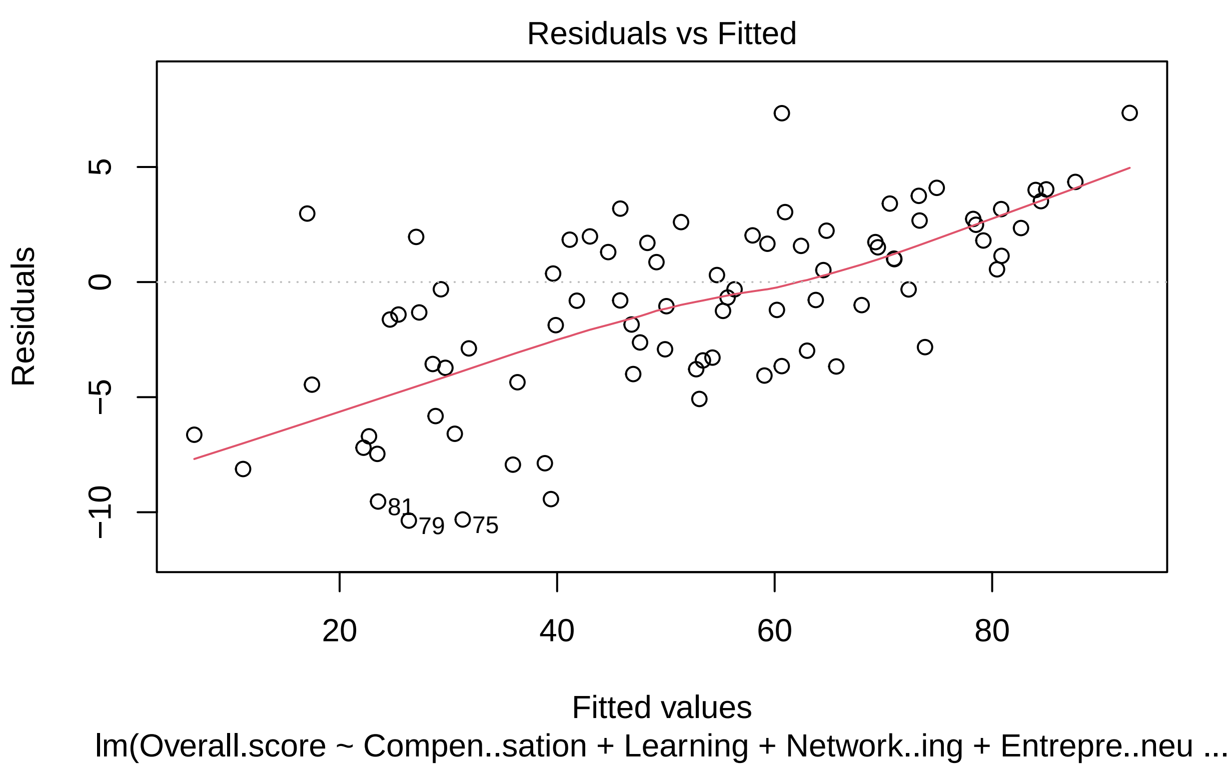
```r
summary(m)
```

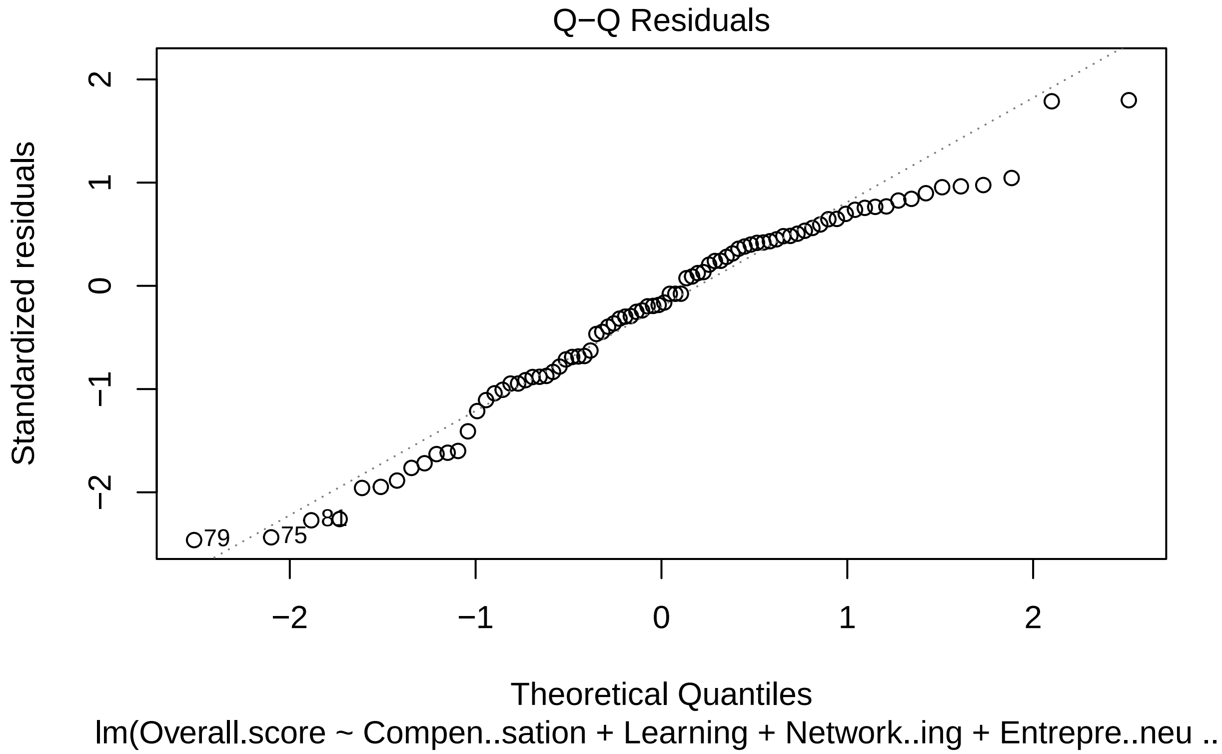
```
##
## Call:
## lm(formula = Overall.score ~ Compen..sation + Learning + Network..ing +
##     Entrepre..neurship + Diversity - 1, data = cr_bloomberg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3634  -3.6558  -0.7267   1.9670   7.3488
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## Compen..sation      0.73090    0.02883  25.354  < 2e-16 ***
## Learning            0.10019    0.03804   2.634 0.010162 *
## Network..ing        0.02436    0.03966   0.614 0.540852
## Entrepre..neurship  0.04766    0.03083   1.546 0.126089
## Diversity           0.06960    0.01757   3.961 0.000162 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.263 on 79 degrees of freedom
## Multiple R-squared:  0.9946, Adjusted R-squared:  0.9943
## F-statistic:  2909 on 5 and 79 DF,  p-value: < 2.2e-16
```

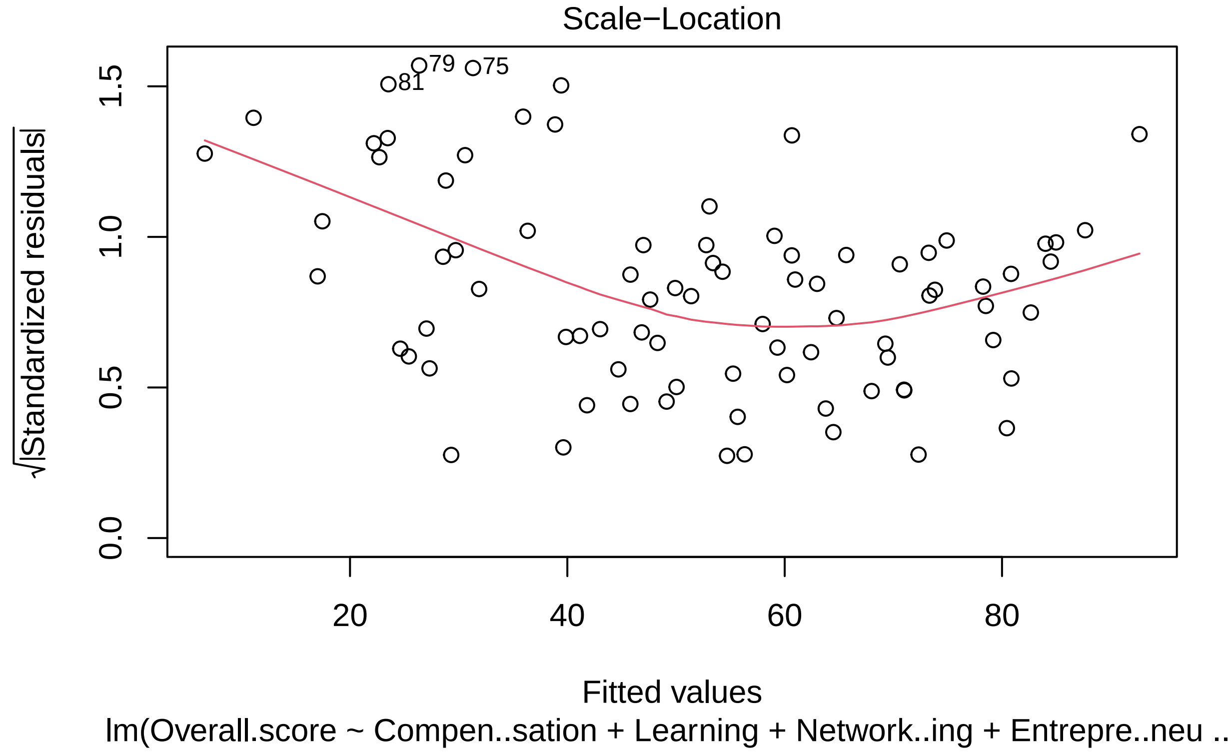
```r
sum(coef(m))
```

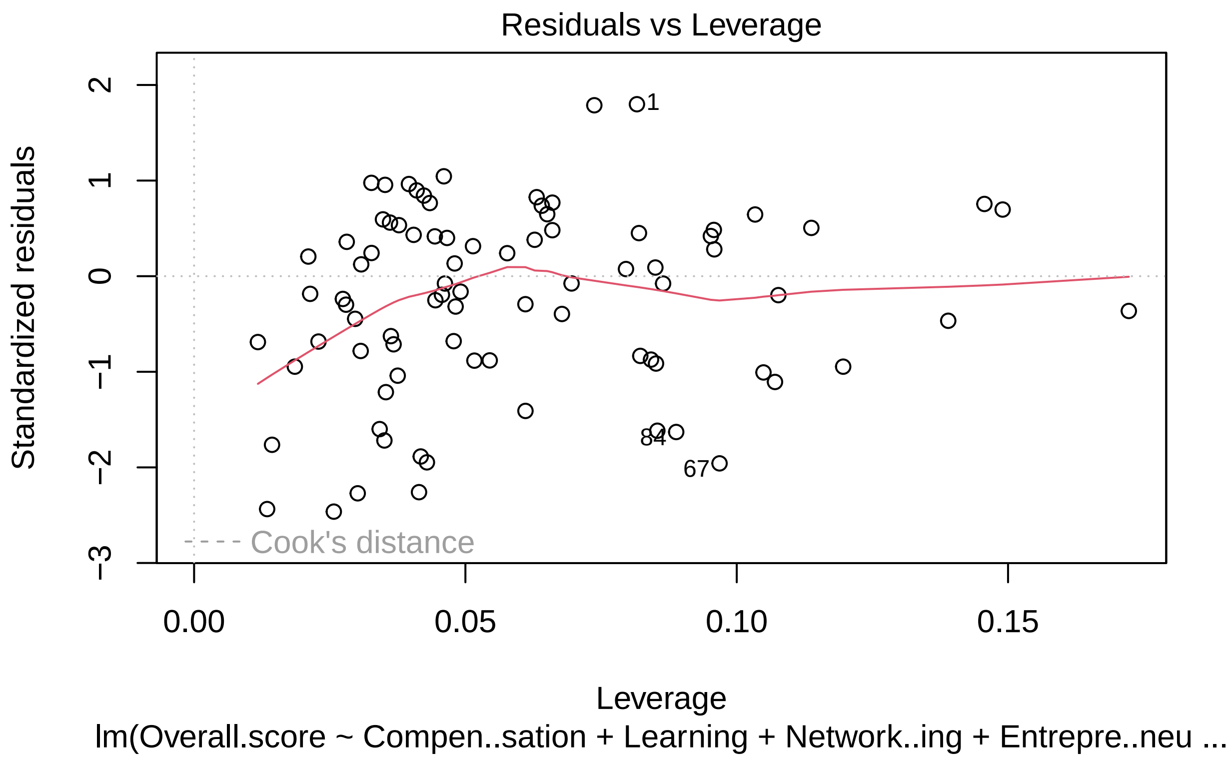
```
## [1] 0.9727114
```

```r
plot(m)
```




Q−Q Residuals
Standardized residuals
−2
−1
0
1
2
79
75
81
Theoretical Quantiles
lm(Overall.score ~ Compen..sation + Learning + Network..ing + Entrepre..neu ...


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
81
79
75
20
40
60
80
Fitted values
lm(Overall.score ~ Compen..sation + Learning + Network..ing + Entrepre..neu ...



my calculations

```
m2 <- lm(formula = my_overall_scaled ~ Overall.score + Learning +
                                  Network..ing + Entrepre..neurship + Diversity - 1, data = cr_bloomberg
summary(m)
```

```
##
## Call:
## lm(formula = Overall.score ~ Compen..sation + Learning + Network..ing +
##     Entrepre..neurship + Diversity - 1, data = cr_bloomberg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
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